

Fat intake modulates cerebral blood flow in homeostatic and gustatory brain areas in humans.



BACKGROUND: The hypothalamus is the central homeostatic control region of the brain and, therefore, highly influenced by nutrients such as glucose and fat. Immediate and prolonged homeostatic effects of glucose ingestion have been well characterized. However, studies that used stimulation with fat have mainly investigated immediate perceptual processes. Besides homeostatic processes, the gustatory cortex, including parts of the insular cortex, is crucial for the processing of food items. **OBJECTIVE:** The aim of this study was to investigate the effect of high- compared with low-fat meals on the hypothalamus and the insular cortex. **DESIGN:** Eleven healthy men participated in a single-blinded, functional MRI study of high- and low-fat meals on 2 measurement days. Cerebral blood flow (CBF) was measured before and 30 and 120 min after intake of high- and low-fat yogurts. Hunger was rated and blood samples were taken before each CBF measurement. **RESULTS:** High-fat yogurt induced a pronounced decrease in CBF in the hypothalamus, and the corresponding CBF change correlated positively with the insulin change. Furthermore, insular activity increased after 120 min in the low-fat condition only. The CBF change in both regions correlated positively in the high-fat condition. **CONCLUSIONS:** The decrease in hypothalamic activity and the interaction with the insular cortex elicited by fat may contribute to an efficient energy homeostasis. Therefore, fat might be a modulator of homeostatic and gustatory brain regions and their interaction. This trial was registered at clinicaltrials.gov as NCT01516021.